

Example 13: Is the following argument valid?
If you do every problem in this book, then you will learn discrete mathematics. You learned discrete mathematics.

Therefore, you did every problem in this book.

Soln: It's easy to see that this is an incorrect argument using the fallacy of affirming the conclusion. Indeed, it is possible for you to learn discrete mathematics in some way other than by doing every problem in this book. (You may learn discrete mathematics by reading, listening to lectures, doing some, but not all, the problems in this book, and so on.) $\square$

The proposition $[(p \rightarrow q) \wedge p] \rightarrow q$ is not a tautology, because it is false when p is false and q is true. Many incorrect arguments use this as a rule of inference. This type of incorrect reasoning is called the Fallacy of denying the hypothesis.

1.6.8 Rules of Inference for Quantified Statements

Universal instantiation: is the rule of inference used to conclude that $P(c)$ is true, where c is ~~one~~ a particular member of the domain, given the premise $\forall x P(x)$. Universal instantiation is used when we conclude from the statement "All women are wise" that "Lisa is wise," where Lisa is a member of the domain of all women.

Universal generalization: is the rule of inference that states that $\forall x P(x)$ is true, given the premise that $P(c)$ is true for all elements c in the domain. Universal generalization is ~~used~~ used when we show that $\forall x P(x)$ is true by taking an arbitrary element c from the domain and showing that $P(c)$ is true. The element that we select must be an arbitrary and not a specific element of the domain. That is, when we assert from $\forall x P(x)$ the existence of an element c in the domain,